

Response to the Second-Round Review

BayesianDEB: A Bayesian Framework for Dynamic Energy Budget Modelling in R

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on behalf of the authors

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We thank the editor and reviewer for the careful second reading of *BayesianDEB: A Bayesian Framework for Dynamic Energy Budget Modelling in R*. Below we reproduce each point of the review of **22 May 2026** (in italics), followed by our response and the specific files in which the change was made. All work refers to package version **0.2.1**; the replication archive is `BayesianDEB_replication.zip` and the revised manuscript is `bayesiandeb.tex`.

A one-line summary of the changes:

- ▶ a single master replication script (`replicate_all.R`) reproduces every manuscript number in about one minute from cached draws;
- ▶ every code listing shown in the manuscript now executes in the replication scripts and prints its output;
- ▶ all manuscript outputs are produced with a fixed seed from the shipped cache, so they reproduce bit-identically;
- ▶ the diagnostics `print` and `plot` methods are now compact;
- ▶ the `bdeb_diagnose` example family moved from `\dontrun{}` to a `cmdstanr`-gated `\donttest{}` block;
- ▶ the dose-response figure now draws the reference lines described in its caption;
- ▶ the README installation instructions no longer pull the stale CRAN version.

1 Package version must match the CRAN version

“The submitted package version is not in line with the package published on CRAN. We require that both versions match.”

Response. CRAN currently holds 0.1.4, whereas the manuscript and this submission describe 0.2.1. We have submitted 0.2.1 to CRAN so that the published package matches the manuscript; the accompanying `cran-comments.md` documents the version bump. `DESCRIPTION` declares `Version: 0.2.1`. The package passes `R CMD check --as-cran` with 0 errors / 0 warnings / 1 note (the note is the informational `cmdstanr-in-Additional_repositories` line discussed below).

2 README should include cmdstanr installation instructions

“The README file should also include instructions for the installation of `cmdstanr`.”

Response. Added. `README.md` (Installation section) now gives the two-step procedure explicitly: install `cmdstanr` from the Stan r-universe and then build the `CmdStan` toolchain once,

```
install.packages("cmdstanr",  
  repos = c("https://stan-dev.r-universe.dev", getOption("repos")))  
cmdstanr::install_cmdstan()
```

and states that `bdeb_fit()` checks for the toolchain at runtime.

3 A single replication script that runs within one hour

“Journal of Statistical Software requires that a single replication script that can be run within one hour is provided. ... parts of the results require that the scripts in `sbc` are run and these scripts are supposed to take more than 20 hours to finish. We would thus ask that the authors arrange their replication scripts into a smaller number of R files and that they also work on reducing the computational time needed to replicate results.”

Response. We have added a single master script, `replication/replicate_all.R`, that reproduces *every figure, table and printed number* in the manuscript with one command (`Rscript replicate_all.R`). By default it does *not* run any MCMC: it loads the archived posterior draws shipped in `outputs/` and `sbc/` and regenerates all outputs in **about one minute** on a laptop — well inside the one-hour budget. The long simulation-based-calibration runs are never executed by this script; their archived rank matrices (`sbc/*.rds`) reproduce Figures 3–4 and Tables 4–5 directly, and regenerating them from scratch (45 min to 12 h per model) is documented separately in `sbc/README.md` and is not part of the replication budget. For readers who wish to recompute, a full refit is available via `BDEB_RECOMPUTE=true BDEB_MODE=full` (~3 h) and a quick refit via `BDEB_MODE=lite` (~30 min).

Reference. `replication/replicate_all.R`, `replication/README.md`, `replication/sbc/README.md`.

4 `example("bdeb_diagnose")` should be executable (`\donttest`)

“`example("bdeb_diagnose")` shows that this example is in `\dontrun`. This should, in principle, be executable code and we expect it to be in `donttest` instead so as to allow its execution using `example`, similarly to what has been implemented for `bdeb_fit`.”

Response. Done. `bdeb_diagnose()` and the `bdeb_diagnostics print/summary/plot` methods — together with `bdeb_loo()`, `bdeb_derived()` and the deprecated `bdeb_summary()` — now wrap their examples in a `cmdstanr`-gated `\donttest{}` block of the same form used for `bdeb_fit()`:

```
\donttest{
if (requireNamespace("cmdstanr", quietly = TRUE) &&
    nzchar(tryCatch(cmdstanr::cmdstan_path(), error = function(e) ""))) {
  data(eisenia_growth)
  dat <- bdeb_data(growth = eisenia_growth[eisenia_growth$id == 1, ])
  fit <- bdeb_fit(bdeb_model(dat, type = "individual"),
                 chains = 2, iter_warmup = 200, iter_sampling = 200,
                 refresh = 0)
  print(bdeb_diagnose(fit))
}
}
```

so `example("bdeb_diagnose")` runs whenever a `CmdStan` toolchain is present. No exported function now uses `\dontrun{}`.

Reference. `R/diagnostics.R`, `R/fit.R`, `R/ppc.R`, `R/debttox.R`.

5 The plot method for `bdeb_diagnostics` could be more readable

“We believe that the plot method for `bdeb_diagnostics` could be improved to allow a shorter and more readable display on screen.”

Response. `plot.bdeb_diagnostics()` now hides the per-time-point latent states (`x_sol[i,j]`, `L_hat[i]`, ...) by default and plots only the scalar model parameters, so the R-hat / ESS panels are short and legible instead of carrying dozens of latent rows. A new `full = TRUE` argument restores the complete plot, and the subtitle reports how many latent rows were hidden. This mirrors the behaviour already adopted for `print.bdeb_diagnostics()`.

Reference. `R/diagnostics.R (plot.bdeb_diagnostics)`; `NEWS.md`.

6 Replication output differs from the article (LOO comparison)

“Results obtained with the replication material (full version) are still different from the results shown in the article. For instance, on page 13, the `loo_compare` for the lognormal model differs.”

Response. The LOO comparison on this individual is genuinely Monte-Carlo sensitive: with only $n = 13$ observations a few Pareto- $\hat{k}$ values exceed 0.7, so the importance-sampling ELPD difference fluctuates by a fraction of its standard error between runs (we observed values in the range -0.7 to -1.1 , all with $SE \approx 1.2$). We have:

- ▶ fixed a global seed (`seed = 42`) in every fit so the shipped result is deterministic;
- ▶ made `replicate_all.R` reproduce the manuscript numbers from the cached draws, so the printed ΔELPD matches the article exactly;
- ▶ reframed the conclusion in the text as qualitative: because $|\Delta\text{ELPD}|$ is smaller than its standard error, neither observation model is decisively preferred, and the exact difference should be read as indicative rather than definitive.

The manuscript now prints `lognormal -1.0 (SE 1.3)` and states plainly that this difference is within its standard error.

Reference. `replication/00_setup.R (seed)`, `replication/01_illustrations.R`, manuscript Section 5.1.

7 README installation pulls the old CRAN version (0.1.4)

“The instructions in the `README.md` suggest installing with a `repos` option that includes CRAN, which installs version 0.1.4 and not the submitted package version 0.2.0.”

Response. The installation instructions have been separated. `BayesianDEB` is now installed with a plain `install.packages("BayesianDEB")` (which will resolve to 0.2.1 once it is on CRAN), and `cmdstanr` is installed from the Stan r-universe in a *separate* call (see Section 2). The combined `repos = c("https://stan-dev.r-universe.dev", getOption("repos"))` command that could shadow the CRAN version of `BayesianDEB` has been removed from the `README`.

Reference. `README.md` (Installation section).

8 Replication material is incomplete (missing manuscript code)

“Some code included in the manuscript which should be executable code is not contained ... E.g., on page 7, `prior_species("Daphnia_magna", type = "debtox")`.”

Response. Every code listing shown in the manuscript is now executed verbatim in the replication scripts. In particular, `prior_species("Daphnia_magna", type = "debttox")` is called and printed in `replication/01_illustrations.R`, alongside the `prior_species("Eisenia_fetida")` listing. We re-audited the manuscript listing by listing and confirmed each one appears in a replication script.

Reference. `replication/01_illustrations.R`.

9 `bdeb_diagnose(fit)` output differs and is too long

“Some results obtained differ. E.g., compared to page 12, `bdeb_diagnose(fit)` reports a different number of divergent transitions and prints a very long table including `x_sol[i,j]` and `L_hat[i]` rows.”

Response. Two changes address this:

- **Reproducibility.** With the fixed seed and cached draws (Section 6), the page-12 output is reproduced exactly by `replicate_all.R`; the manuscript now reports a single divergent transition (1 of 4000 draws, < 0.1%) and discusses it.
- **Length.** `print.bdeb_diagnostics()` now shows only the scalar model parameters by default and hides the per-time-point latent states; the manuscript output on p. 12 is correspondingly compact (eight parameter rows, no `x_sol/L_hat`). The full table remains available via `print(x, full = TRUE)` or `summary(x)$table`.

Reference. `R/diagnostics.R`, `replication/01_illustrations.R`, manuscript Section 5.1.

10 `bdeb_derived()` result is created but never printed

“The replication material creates `der <- bdeb_derived(fit, ...)` but does not print it, so it is not easy to compare with the manuscript.”

Response. Fixed. `replication/01_illustrations.R` now prints the derived-quantity table immediately after constructing it, in the same layout as the manuscript:

```
der <- bdeb_derived(fit,
  quantities = c("L_m", "L_inf", "k_M", "g", "growth_rate"), f = 1.0)
print(as.data.frame(summarise_draws(der, "mean", "sd",
  "q5" = ~quantile(.x, 0.05), "q95" = ~quantile(.x, 0.95))), digits = 3)
```

Reference. `replication/01_illustrations.R`.

11 LOO triggers Pareto- $\hat{k}$ warnings that are not commented on

“Warnings are triggered (Some Pareto k diagnostic values are too high) where one might also point them out and comment on them.”

Response. These warnings are now explicitly anticipated and explained. In `replication/01_illustrations.R` a comment precedes the `loo_compare()` call noting that, with $n = 13$, a few $\hat{k} > 0.7$ are expected and that the importance-sampling ELPD is therefore indicative. The manuscript (Section 5.1) makes the same point in the running text: the Pareto- $\hat{k}$ warning is reported, interpreted (the approximation is unreliable for a few points), and used to justify reading the comparison qualitatively.

Reference. `replication/01_illustrations.R`, manuscript Section 5.1.

12 Figure caption refers to lines that are not in the figure

“The caption of Figure 2 discusses something dashed red and dotted green. However, it is unclear to what in the figure one refers.”

Response. `plot_dose_response()` now actually draws these reference lines: a dashed red vertical line at the posterior-median EC_{50} and a dotted green vertical line at the posterior-median NEC, with a matching legend subtitle. Non-finite draws are dropped and the view is clipped with `coord_cartesian()` so degenerate draws no longer create vertical artefacts. The Figure 2 caption now matches the rendered figure element by element.

Reference. `R/debtox.R (plot_dose_response)`, manuscript Figure 2 caption.

13 Unclear which part of the replication reproduces Figures 3 and 4

“It is unclear what part of the replication material reproduces Figures 3 and 4.”

Response. `replication/README.md` now contains an explicit *Figure* $\rightarrow$ *script* table mapping every manuscript figure to the script and cached object that produces it. Figures 3 and 4 are listed as: Fig. 3 (`fig_sbc_ranks.pdf`) produced by `02_validation.R` from `sbc/sbc_results.rds`; Fig. 4 (`fig_sbc_hierarchical.pdf`) produced by `02_validation.R` from `sbc/sbc_hierarchical_results.rds`. The same table maps all remaining figures and the corresponding manuscript sections.

Reference. `replication/README.md` (Figure $\rightarrow$ script and Section $\rightarrow$ script tables).

We thank the editor and reviewer again for the constructive feedback, which has further improved the package and its replication material.

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